

# Written Report Requirements

## Purpose

The written report demonstrates that you understand the complete project, not only that you can run code. Explain what you did, why you did it, what the model found, and where the results are limited.

## Required format

Submit either a Markdown file or a PDF exported from your Markdown document. Use clear headings, complete sentences, readable tables, and captions for all figures. Write in your own words.

Recommended length: approximately 1,000–1,500 words, excluding code and figure captions.

## Required sections

| Section              | Required content                                                                                                  |
|----------------------|-------------------------------------------------------------------------------------------------------------------|
| Title                | Project title, student name, and date                                                                             |
| Problem definition   | What is being predicted, why it is regression, and the project question                                           |
| Dataset              | Number of rows, columns, target, predictors, numerical fields, categorical fields, and synthetic-data disclaimer  |
| Data preparation     | Missing-value counts, duplicate check, imputation choices, categorical encoding, and any other cleaning decisions |
| Exploratory analysis | Important relationships and patterns supported by tables or visualizations                                        |
| Model                | Train-test split and Linear Regression method                                                                     |
| Evaluation           | MAE, MSE, RMSE, and $R^2$ , with plain-language explanations                                                      |
| Visualizations       | Three required Matplotlib figures with captions and interpretation                                                |

|                 |                                                                                      |
|-----------------|--------------------------------------------------------------------------------------|
| Interpretation  | What the results suggest and where the model makes larger errors                     |
| Limitations     | At least three limitations, including synthetic data and model limitations           |
| Conclusion      | Direct answer to the project question and key learning points                        |
| Reproducibility | Libraries, dataset location, random state, and instructions for running the notebook |

## Required visualizations

Include these three figures:

1. House area versus sale price with a fitted line or clearly explained relationship.
2. Actual versus predicted test-set sale prices with a perfect-prediction reference line.
3. Residuals or prediction errors with labeled axes and an explanation.

Figures must be generated with Matplotlib. Do not submit charts without titles, axis labels, or captions.

## Evaluation guidance

Report the metric values produced by your own run. Do not copy values from another student. Explain that MAE and RMSE are in price units, while  $R^2$  summarizes the proportion of variation explained by the model on the test data.

## Academic integrity

You may learn from documentation and tutorials, but the submitted analysis and explanations must be your own. You must understand and be able to explain the code you submit.